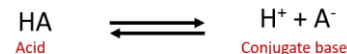


Inorganic chemistry

Acids and bases

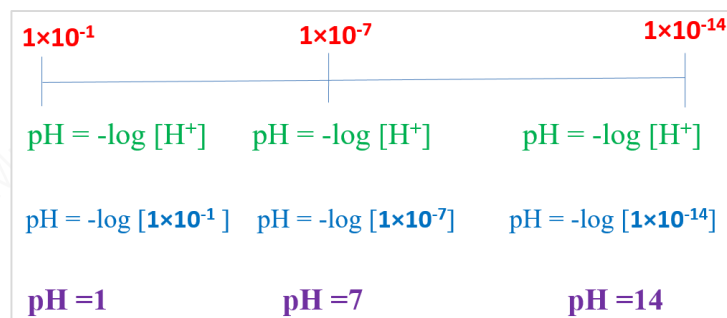
- **Acids** are molecules which can **donate** a proton (hydrogen ion, H^+).
- **Bases** are proton **acceptors**
- Strength of an acid depends on how readily it does that proton donation
- Measured by the **acid dissociation constant K_a**



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

What is pH?

- Measurement of the amount of protons in a solution
- Logarithm to base 10 of the proton concentration
- An equivalent measure for acid strength is: **$pK_a = -\log_{10} [K_a]$**
- **$pH = -\log [H^+]$**
 - A solution is **acidic** if its pH is less than 7
 - A solution is **basic (alkali)** if its pH is more than 7
 - A solution is **neutral** if its pH is equal to 7
- **pH paper** changes color to indicate a specific pH value.



Buffers

- A solution consists of a **weak acid and its salt** or a **weak base and its salt**.
- A solution that **resists change in pH** when limited amounts of an acid or base are added to it.
- pH the closer the pH of the buffer is to the pK_a of the weak acid the greater the buffer capacity.
- Concentration: the greater the concentration of the weak acid and its conjugate base, the greater the buffer capacity.

Blood buffers

- The normal range pH of human blood is between **(7.35 - 7.45)**.
- Any change **longer than 0.1 pH unit** in either direction may cause illness.
- If the pH goes below 6.8 or above 7.8 death may result.
- To hold the pH of the blood close to 7.4, the body uses three buffer systems: **carbonate**, **phosphate**, and **proteins**.
- The most important of these systems is the **carbonate buffer**.

$$pH = pK_a + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

Acidosis and alkalosis

- If the pH goes lower than 7.35, the condition called **acidosis**.
- A blood pH higher than 7.45 produces **alkalosis**.
- **Acidosis** leads to **depression of the nervous** system, mild acidosis can result to **fainting**, a more severe case can cause **coma**.
- **Alkalosis** leads to **over stimulation of the nervous** system, **muscle cramps**, and **convulsions**.
- If the acidosis or alkalosis persists for a sufficient period of time, or if the pH gets too far away the 7.35-7.45, death may result.

Respiratory acidosis (pH < 7.35 , pCO₂ > 45 mm Hg)

- Results from difficulty in breathing (**hypoventilation**).
- An obstruction in the windpipe or diseases such as **pneumonia**, **emphysema**, **asthma**, or **congestion heart failure** may diminish the amount of CO₂ that leaves the body through the lungs. (you can produce mild acidosis by holding your breath).
- The pH of the blood decreases because the CO₂ unable to escape fast enough, remain in the blood. where it lowers the **[HCO₃⁻] / [H₂CO₃]**. ($\text{CO}_2 + \text{O}_2 \longrightarrow \text{H}_2\text{CO}_3$)
- **Cause:** Respiratory failure
- **Symptoms:** coma
- **Therapy:** Therapy of underlying disease , Mechanical ventilation

Respiratory alkalosis (pH > 7.45 , pCO₂ < 35 mm Hg)

- Involves the reverse process, it arises from rapid or heavy breathing, called hyperventilation.
- Respiratory alkalosis can come about from **fever**, **infection**, the **action of certain drugs** or even **hysteria**.
- Here the excessive loss of CO₂, raises the **[HCO₃⁻] / [H₂CO₃]** ratio and the pH.
- **Causes:** Hypoxic respiratory failure, Salicylate intoxication, Early sepsis, Hepatic failure, and artificial hyperventilation
- **Symptoms:** Decreased cerebral or coronary blood flow, Neuromuscular excitability
- **Therapy:** Therapy of underlying disease, Increasing the dead space

Metabolic acidosis

- The causes of metabolic acidosis are **starvation (fasting)**, and **heavy exercise**.
- When the body doesn't get enough food, it burns its own fat, and the products of this reaction are acidic compounds, that enter the blood. This problem sometimes arises in people on fad diets.

- Heavy exercise causes the muscles to produce excessive amounts of **lactate**, which makes muscles feel tired and sore.
- METABOLIC ACIDOSIS is also caused by **a number of metabolic irregularities**. For example, the disease of diabetes mellitus produces acidic compounds called **ketone bodies**.
- **Symptoms**: Increased sympathetic activity, decreased enotropy, arterial vasodilatation – critical pH (7.2), Decreased oxy-Hgb binding, Hyperkalaemia, Insulin resistance, Free radical formation, Bone demineralization, Hyperventilation.

Metabolic alkalosis

- Can result from various metabolic irregularities as well. In addition, it can be caused by **excessive vomiting**.
- The contents of the stomach are strongly acidic (pH 1-3), and loss of substantial amounts of this acidic material raises the pH of the blood.
- **Symptoms**: Hypoventilation, respiratory depression, Neuromuscular excitability, Hypokalaemia, Seizures, increased oxy-Hgb affinity, altered coronary blood flow, vasoconstriction, Decreased cerebral blood flow.

	pH	pCO ₂	HCO ₃ ⁻	HCO ₃ / H ₂ CO ₃
Normal	7.35 - 7.45	40 mmHg	20 mm/Hg	20
Metabolic acidosis	Decreased	Normal	Decreased	Decreased
Metabolic alkalosis	Increased	Decreased	Normal	Increased
Respiratory acidosis	Decreased	Increased	Normal	Decreased
Respiratory alkalosis	Increased	Decreased	Increased	Increased

Acid Rain

- Pollution in the air (sulfur dioxide, carbon dioxide, nitrogen dioxide) combines with water to form various acids. ($\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{CO}_3$)
- Rapid changes in pH can kill fish and other organisms in lakes and streams.
- Soil pH is affected and can kill plants and create sinkholes.

Anion gap

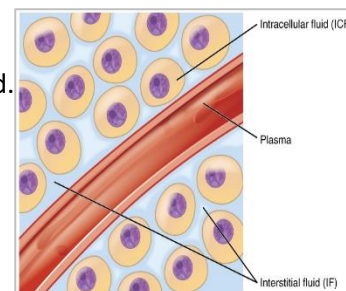
- The sum of cations and anions in extracellular fluid is always equal so as to maintain the electrical neutrality.
- **Cations (95%)** were maintained by (Na and K ions), whereas, (Cl⁻ and HCO₃⁻ ions), **account for (86%)** of anions. There are commonly measured electrolytes and hence there is a difference between cations and anions.
- The difference of cations and anions or the unmeasured anions constitute the anion gap, which is due to phosphorous, SO₄ ion, PO₄ ion, and organic salts.
- The difference between (Na + K ions) and (Cl + HCO₃ ions) is normally **about 12±5 mEq/L** (mMols/L). Alteration in anion gap is extremely useful in the clinical assessment with acid-base disorders.

Body Fluids ✓

- Total amount of **fluid** in the human body is approximately **70%** of body weight.
- In the body **total body water 42L** (**60%** of the body weight)
- Body fluid has been divided into two compartments:
 - **Intracellular fluid (ICF)** \\ **66.7%** (2/3) of total body water. **28 L** (40% of the body weight)
 - **Extracellular fluid (ECF)** \\ **33.3%** (1/3) of total body water. **14 L** (20% of the body weight)
- **Extracellular fluid includes:**
 - **Interstitial fluid**\\
 - Approximately **80%** of extracellular fluid (ECF)
 - **Plasma**\\
 - Approximately **20%** of extracellular fluid (ECF)
 - Extracellular fluid (ECF) also includes: Lymph, synovial fluid, aqueous humor (in eyes), cerebrospinal fluid, vitreous body, endolymph, perilymph, pleural, pericardial and peritoneal fluids.

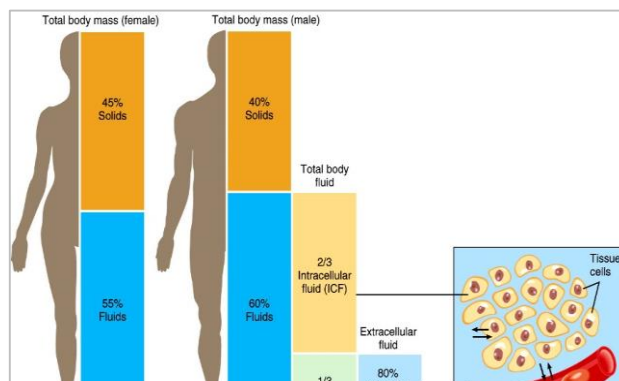
Barriers separate intracellular fluid (ICF), interstitial fluid and plasma:

- **Plasma membrane**\\ Separates intracellular fluid (ICF) from surrounding interstitial fluid.
- **Blood vessel wall**\\ Separate interstitial fluid from plasma.



Intracellular fluid (ICF)	Extracellular fluid (ECF)
• Most abundant <u>cation</u> is K⁺ (Potassium) • <u>Anion</u> are proteins and phosphates (HPO_4^{2-})	• Most abundant <u>cation</u> is Na⁺ (Sodium) • Most abundant <u>anion</u> is Cl⁻ (Chloride)

- **Na⁺ /K⁺ pumps** play major role in keeping K⁺ high inside cells and Na⁺ high outside cell.
- In adult males, total body water ranges between 55% and 65% of body weight depending on the proportion of body fat. Adult women are about 10% less than adult men. New born infants have approximately 75% body water.
- On weight basis, 32 ml/kg for adults and 100-150 ml/kg for infants.



Composition of body fluids:

- **Organic substances** \\ Glucose, amino acids, fatty acids, hormones, and enzymes.
- **Inorganic substances** \\ Sodium, Potassium, Calcium, Magnesium, Chloride, Phosphate, and Sulphate.

Sodium Na^+

- **Most abundant cation in extracellular fluid (ECF)**
- **90%** of extracellular cations
- **Muscle contraction**
- **Impulse transmission**
- Plays pivotal role in **fluid and electrolyte balance** as it accounts for half of the osmolarity of ECF.

Chloride Cl^-

- Most prevalent **anion** in extracellular fluid (ECF)
- Moves easily between ECF and ICF because most plasma membranes contain Cl^- leakage channels and transporters.
- Regulates **osmotic pressure**
- **Forms HCl** in gastric acid
- Can help balance levels of anions in different fluids.

Bicarbonate HCO_3^-

- Second most prevalent extracellular anion.
- Concentration **increases in blood passing** through systemic capillaries picking up carbon dioxide.
- Chloride shift helps maintain correct balance of anions in ECF and ICF.

Potassium K^+

- **Most abundant cation in intracellular fluid (ICF).**
- Establish **resting membrane potential** in neuron and muscle fibers.
- **Action potentials**
- Maintains normal **ICF fluid volume.**
- Helps **regulate pH** of body fluids when exchanged for H^+ .

Magnesium

- Mg^{2+} is second most common intracellular cation.
- Cofactor for certain enzymes and sodium-potassium pump
- Essential for **synaptic transmission, normal neuromuscular activity and myocardial function.**

Specialized Fluids of the Body:

- Lymph, Milk, Cerebrospinal fluid, Amniotic fluid, Aqueous humor, Sweat, and Tears.

Lymph

- Clear and colorless fluid.
- 96% water and 4% solids.
- **Solids:**
 - **Proteins (2-6% of solids)** \\ albumin, globulin, fibrinogen, prothrombin, clotting factors, antibodies, enzymes.
 - **Lipids (5-15% of solids)** \\ Chylomicrons and lipoproteins
 - **Carbohydrates** \\ Glucose mainly
 - **Non-protein nitrogen (NPN)** \\ Urea and creatinine
 - **Electrolytes** \\ Sodium, calcium, potassium, chloride, bicarbonates
- **Functions of Lymph:**
 - Return protein from tissue spaces into blood.
 - Redistribution of fluid.
 - Removal of bacteria, toxins and other foreign bodies from tissues.
 - **Maintain** structural and functional integrity of tissue.
 - Route for **intestinal fat absorption**.
 - Transport lymphocytes.

Milk

- Complete natural food
- 83-87% water and 13-17% solids
- **Solids:**
 - **Carbohydrates** \\ Lactose
 - **Lipids** \\ Triacylglycerols, Palmitic acid, myristic acid, stearic acid, lauric acid, butyric acid, and oleic acid.
 - **Proteins** \\ Casein (80%), Lactalbumin, Enzymes (proteases, xanthine oxidase, lysozyme), immunoglobulins.
 - **Minerals** \\ Ca, Mg, P, Na, K, Cl.
 - **Vitamins** \\ Fat and water soluble vitamins except Vit-C.
- **Functions of Milk:**
 - Milk sugar provides galactose, a structural unit for growing infant.
 - In intestine, it gets metabolized to lactic acid which eliminates harmful bacteria.

Cerebrospinal fluid (CSF)

- Clear, colorless liquid formed within the **cavities of brain** and **around spinal cord**.
- At any given time, there is 120-150 ml Cerebrospinal fluid (CSF) in the system.
- Cerebrospinal fluid (CSF) is completely replaced about three times a day.
- **Functions of Cerebrospinal fluid (CSF):**
 - Hydrolic shock absorber
 - Regulation of intracranial pressure
 - Influences the hunger sensation and eating behaviours.

Amniotic Fluid

- Liquid produced by membranes and fetus
- Volume of fluid increases with gestational age
- Clear with some desquamated fetal cell and a little lipid.
- **Functions of Amniotic Fluid (AF):**
 - **Physical protection** to the fetus
 - Medium **for exchange** of various chemical

Aqueous Humor

- Fluid that fills the interior chamber of eye
- **Secreted by ciliary body**, enters anterior chamber
- Blockade in the flow of aqueous humor causes **glaucoma** due to increased intraocular pressure.
- Posterior chamber of eye is filled with **vitrous** humor which contains a gel (vitrous body of hyaluronic acid secreted by retina)

Sweat

- **Regulates body temperature** by cooling and evaporation.
- Sweat glands controlled by **ANS, Adrenal cortical steroid** - which affect the quantity of electrolyte present
- Insensible perspiration amounts to 800-1200ml/day
- Volume of sweat produced/day during muscular exercise at elevated temperature may range from 10-14l, which may lead to water and electrolyte imbalance
- Water content of sweat varies from 99.2-99.7%
- pH – **4.7 to 7.5**
- Total non-protein nitrogen (urea) varies from 0.07-1% per hour during copius sweating
- Electrolyte in sweat varies: (Na^+ 12.6-127 meq/l , K^+ 5-32 meq/l , Cl^- 8.5-85 meq/l)

Tears

- Produced by lachrymal glands
- Isotonic but becomes hypertonic due to evaporation as fluid passes over the cornea
- When the tear flow is copius, fluid is isotonic
- Under stimulus with a slow rate of tear flow, the fluid is about 25m osm hypertonic

Copius – Rapid tear flow induction

- pH – **7 to 7.6** due to loss of CO_2
- Protein content is 0.6 to 0.18g/dl with an **albumin/globulin ratio of 1:5 or 2:1**
- Small amount of mucin is also present
- **Lysozyme** – lyses the cells of a number of micro-organisms by breaking down the polysaccharides of their outer layer
- Lysozyme protects eye from infectious agents
- Lubricate the surface of the cornea
- Fill the irregularities of the corneal surface to improve optical properties
- Protects eyes from injury.

Osmolarity (osmotic concentration)

- Is the measure of **solute concentration**, defined as the number of **osmoles** (Osm) of solute per **litre** (L) of **solution** (osmol/L).
- The osmolarity of a solution is usually expressed as **Osm/L** (pronounced "osmolar"), in the same way that the **molarity** of a solution is expressed as "M" (pronounced "molar"). Whereas molarity measures the number of **moles** of solute per unit **volume** litre(L) of solution.
- This value allows the measurement of the **osmotic pressure** of a solution and the determination of how the solvent will diffuse across a **semipermeable membrane** (**osmosis**) separating two solutions of different osmotic concentration.
- The unit of osmotic concentration is the **osmole**. This is unit of measurement that defines the number of moles of solute that contribute to the osmotic pressure of a solution.
- A **milliosmole (mOsm)** is 1/1,000 of an osmole.
- A **microsmole (μOsm)** (also spelled **micro-osmole**) is 1/1,000,000 of an osmole.
- **Osmolarity is distinct from molarity** because it measures osmoles of solute particles rather than moles of solute. The distinction arises because some compounds can **dissociate** in solution, whereas others cannot.
- **Ionic compounds** such as **salts**, can dissociate in solution into their constituent **ions**, so there is not a one-to-one relationship between the molarity and the osmolarity of a solution.
- **For example**, **sodium chloride** (NaCl) dissociates into Na^+ and Cl^- ions. Thus, for every **1 mole of NaCl** in solution, there are **2 osmoles** of solute particles (1 mol/L NaCl solution is a 2 osmol/L NaCl solution). Both **sodium and chloride** ions affect the **osmotic pressure** of the solution.
- Another **example** is **magnesium chloride** (MgCl_2), which dissociates into Mg^{2+} and 2Cl^- ions. For every 1 mole of MgCl_2 in the solution, there are **3 osmoles** of solute particles.
- **Non-ionic compounds do not dissociate**, and form only 1 osmole of solute per 1 mole of solute.
- **For example**, a 1 mol/L solution of **glucose** is **1 osmol/L**.

Plasma osmolarity vs. osmolality

- Plasma osmolarity can be calculated from **plasma osmolality** by the following equation:
- **Osmolality**: defined as the number of **osmoles** (Osm) of solute per **kilogram** of **solution** (osmol/Kg).
- **Osmolarity** = **osmolality** $\times (\rho_{\text{sol}})$ *→ density of solution*
- where: ρ_{sol} is the **density** of the solution in g/ml, which is **1.025 g/ml** for **blood plasma**.

g/ml 1.025 g/ml

Osmolarity (osmotic Concentration)

- Proportional to the number of osmotic particles formed: **Osm/L = moles $\times$ n** (n, # of particles in solution)
- Assuming complete dissociation:
 - 1 mole of NaCl forms a 2 osmolar solution in 1L
 - 1 mole of CaCl₂ forms a 3 osmolar solution in 1L

Milliequivalents, millimoles and milliosmoles

- **Osmolarity** is the milliosmoles of solute per liter of solution.
- **Osmolality** is the milliosmoles of solute per kilogram of solvent.
- Normal serum osmolality is considered within the range of **275 to 300 mOsmol/kg**.
- **Osmometers** are commercially available for use in the laboratory to measure osmolality
- Abnormal blood **osmolality** can occur with shock, trauma, burns, electrolyte imbalance, hyperglycemia or renal failure.

Clinical considerations of water and electrolyte balance

- Fluid and electrolyte therapy (oral or parenteral) is provided either for maintenance requirements or to replace serious losses or deficits.
- **Example**, a patient taking **diuretics** may simply **require a daily oral potassium supplement** along with adequate intake of water. Hospitalized patients commonly receive parenteral therapy of fluids and electrolytes to support ordinary metabolic functions.
- **Osmolality** of intracellular fluid and extracellular fluids is about equal (cell membrane) this value is about **290 mOsm/kg water**.
- The plasma osmolality is approximated by the formula:

$$\text{plasma osmolality (mOsm / kg)} = 2([Na] + [K])_{\text{plasma}} + \frac{[BUN]}{2.8} + \frac{[Glucose]}{18}$$

- Where Na and K are in mEq/L and **blood urea nitrogen (BUN)** and glucose concentrations are in mg/100 ml (mg/dl)
- If the patient has ingested **ethanol**, the ethanol level should be included in the calculated osmolality:

$$\text{Osmolality} = 2 \{[Na^+] + [K^+]\} + [BUN]/2.8 + [Glucose]/18 + [Ethanol]/3.7$$

- NOTE: Based on the molecular weight of ethanol the divisor should be 4.6 but empiric data shows that ethanol does not behave as an ideal osmole.

Tonicity

Isotonic solutions:

- Are two solutions have the same solute concentration
- The normal salt concentration of the blood is approximately equal to that of 0.9% NaCl solution. The common name for 0.9% NaCl solution is physiologic saline solution. The blood and physiologic saline solution are isotonic they have the same conc.
- 5.5% glucose solution is also approximately isotonic with body fluids.

0.9% NaCl vs blood
5.5% glucose vs body fluids

Hypotonic solutions:

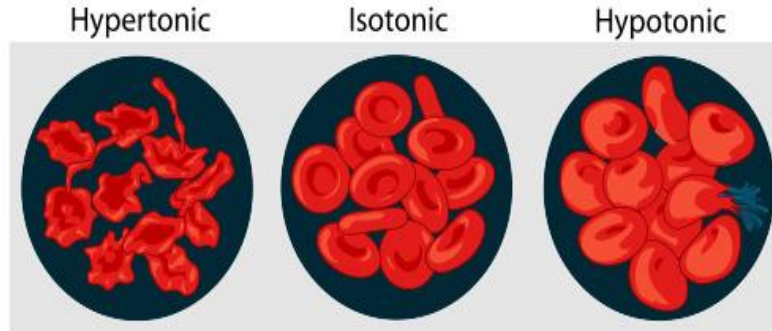
- Hypotonic solution is one that contains lower solute concentration than that of another solution.
- Distilled water and tap water are hypotonic compared with blood.

low conc.

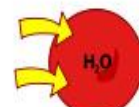
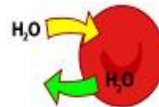
0.9% NaCl
high conc.

Hypertonic solutions:

- Hypertonic solution is one that contains higher solute 5% of NaCl solution or 10% glucose solution are examples of hypertonic solutions, when compared with blood.



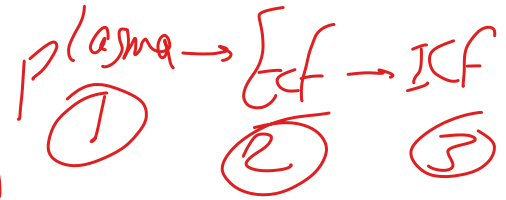
c. retraction



hemolysis

- Isotonic: osmolarity = 0.3 osm/l (NaCl 0.15 mol/l)
- Hypotonic: osmolarity < 0.3 osm/l
- Hypertonic: osmolarity > 0.3 osm/l

Water and minerals



Dehydration (Hypovolemia)

- Loss of water from the body in excess amounts leads to **dehydration**.
- First the plasma becomes concentrated followed by other ECF and later ICF.
- When water comes out of the cells, it also passes into **extracellular fluid (ECF)** in exchange of K & Na passes into ICF from ECF.
- Loss of more than 20% of body water results in death.



Causes of dehydration:

- 1- Severe **diarrhea** and **vomiting**
- 2- **Non-availability of drinking water** as in the case of desert.
- 3- Difficulty in **swallowing** and state of unconsciousness.
- 4- **Loss of fluid from skin** in case of **burns**.
- 5- Diabetic insipidus (ADH deficiency-polyuria)
- 6- **Heart stroke**

Features of dehydration:

- 1- **Dryness of skin, tongue, throat** *increased*
- 2- Changes in the values of packed cell volume (PCV), Hb, plasma protein, plasma electrolytes, urea and **decreased blood volume**

- Treatment of dehydration: Consuming plenty of plain water or water containing sugar and salt depends upon the cause of dehydration. if the condition is very severe, **intravenous infusion of fluids (normal saline)** is required.

Water excess (hydration) hypervolemia:

hydration HAR

Causes of hydration:

- 1- **Hypersecretion of ADH**
- 2- Following the administration of anesthetics its effect takes for about **12-36 hour after the surgery**
- 3- **Renal failure**

Features of hydration:

- 1- Mental confusion, in coordination, muscular weakness, nausea.
- 2- **Decreased** packed cell volume (**PCV**).
- 3- **Decreased plasma electrolytes, plasma osmolality, Increased urine osmolality and increased blood pressure.**

Treatment of hydration:

- 1- **Withdrawal of fluids.**
- 2- Administration of **diuretics**.

Minerals

Minerals are inorganic ions and elements required with the diet for growth, maintenance of tissue and regulation of body processes.

1- Macrominerals

Which are required in amounts in **greater than 100mg/day**, which include; (Calcium, Phosphorus, Sodium, Potassium, Chloride, and Magnesium)

2- microminerals

Which are required in amounts in **less than 100mg/day**, which include; (Iron, Iodine, Manganese, Cobalt, Fluoride, Chromium, Selenium, Zinc and, Copper).

Sodium; [Na⁺]

- Is the major cation of the extra cellular fluid (ECF).
- **Sources**; Milk, cereal, legumes, eggs, vegetables like carrots and tomato, and table salt.
- **Requirement**; 5 gm/day
- **Normal range**; 130-143 Meq/L
- **Functions**:
 - 1- responsible for osmolality of the plasma.
 - 2- plays a main role in maintaining the normal distribution of water and the osmotic pressure in the ECF compartment.
 - 3- regulate the electrolyte and pH balance of the extra cellular compartment.
 - 4- helps in the active transport of glucose, galactose, and amino acids across intestinal mucosa and for Na⁺/K⁺ ATPase.

Hyponatraemia

- Decrease in plasma concentration of [Na ion] may be due to **defect** in kidneys or adrenal cortex.
- Sweating, burns, vomiting, or diarrhea which can be cause loss of [Na ion] containing fluids.
- In adrenal cortex insufficiency; (**Addisons disease**), **decrease** of serum [Na ion] and **increase** in Na ion secretion are seen.
- **Signs and Symptoms**; Confusion, Nausea, and Fatigue

Hypernatraemia

- Increased plasma concentration of [Na ion]. it is due to decrease loss of [Na ion] in body fluids, may occur in decrease activity of ADH, Hyperactivity of the adrenal cortex as in **Cushing's syndrome**.
- Administration of cortisone also leads to increase in serum Na ion.
- Hypertensive patients are advised to take less salt. about 1gm/day.
- **Signs and symptoms**; Lethargy, Restless, Spasticity, and Seize up.

Potassium; [K ion]

- Is the major cation of the intra cellular fluid (ICF).
- **Sources**; vegetables, fruits, and nuts
- **Normal range**; 3.5-5 Meq/L
- **Functions**:
 - 1- Is the principle cation of ICF.
 - 2- [K] ion is required for functioning of nerve, skeletal muscle and cardiac muscles.
 - 3- [K] ion is required as a cofactor in several enzymatic reactions in the body such as pyruvate kinase, glycogen synthesis and protein synthesis.
 - 4- [K] ion is involved in acid-base balance.

Hypokalemia;

- Decrease in plasma concentration of [K ion] This condition **decreases** the heart beat and interferes with vital muscles as those involved in respiration.
- Hypokalemia can occur in any illness, wasting disease, intestinal fistulas and in diarrhea.
- Since aldosterone increases the excretion of K ion. **Cushing syndrome** or administration of cortisone leads to hypokalemia. certain diuretics increase the excretion of K ion.
- Therefore, important to supplement enough K ion when these diuretics are used.
- **Signs and symptoms**; Irregular heartbeat, Loss of appetites, and Muscle cramps

Hyperkalemia;

- Increased plasma concentration of [K ion]. it occurs in **Addison's disease** and in intra venous infusion of K ion at a rate excess of 25 mmol/hour.
- Causes are acute **renal failure**, **chronic renal failure**, **glomerulonephritis**.
- Tissue trauma causes the cells to release K ion into the ECF are burns, traumatic injury and intestinal bleeding.
- **Signs and symptoms**; Fatigue, weakness, tingling, numbness, paralysis, palpitations and difficulty in breathing.

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